MEDICOS MINDS

Better Movement Better Life...

Same People... Different Stories... Same Goal... HEALTH

PHYSIOTHERAPY IN NEUROLOGICAL CONDITIONS

EVIDENCE | FUNCTION | INDEPENDENCE

Restore Function Rebuild Independence

KEY AREAS

- Neuroanatomy & Neurophysiology
- Assessment & Examination
- Goal Setting & Prognosis
- Bobath Concept
- Brunnstrom Approach
- Proprioceptive Neuromuscular Facilitation (PNF)
- Motor Learning & Rehabilitation
- Assistive Devices & Orthotics
- Functional Training
- Community Reintegration

COMMON CONDITIONS

- Stroke
- Parkinson's Disease
- Multiple Sclerosis
- Traumatic Brain Injury (TBI)
- Spinal Cord Injury (SCI)
- Cerebral Palsy
- Peripheral Neuropathy
- Guillain-Barré Syndrome
- Ataxia
- Dementia

Movement is Medicine

ASSESSMENT TOOLS

- Berg Balance Scale
- Fugl-Meyer Assessment
- Modified Ashworth Scale
- Tinetti Balance & Gait
- Functional Independence Measure (FIM)

GOALS

- Improve Mobility
- Enhance Balance
- Increase Strength
- Maximize Independence
- Better Quality of Life

KEY PRINCIPLES

- Neuroplasticity
- Task Specific Training
- Repetition & Intensity
- Functional Goals
- Patient-Centered Approach

TECHNIQUES

- Therapeutic Exercises
- Balance Training
- Gait Training
- Functional Electrical Stimulation (FES)
- Manual Therapy
- Education & Caregiver Training

Small Progress = Big Changes

MOVE | REGAIN | REBUILD | INDEPENDENCE

Progress not Perfection

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PHYSIOTHERAPY IN NEUROLOGICAL CONDITIONS
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HIGH-YIELD BPT STUDY GUIDE & EXAM COMPANION

PT in Neurological conditions

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OFFICIAL DIGITAL STUDY GUIDE VERIFICATION

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STUDY GUIDE DETAILS & CONTENTS:

Comprehensive high-yield notes designed for Bachelor of Physiotherapy (BPT), MBBS, BDS, and allied healthcare students. Covers fundamental anatomy, clinical assessments, pathological conditions, evidence-based physical therapy rehabilitation protocols, and exam-oriented high-yield review.

KEY HIGHLIGHTS:

- Exam-Focused: Concise points curated from leading medical textbooks and clinical references.
- Visual Diagrams: High-yield diagrams and tables for quick revision.
- Mobile & Tablet Optimized: Readable on phones, tablets, iPads, and laptops.
- Official Medicos Minds Release: Verified by Dr. MD Kamran Alam.



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PHYSIOTHERAPY IN NEUROLOGICAL CONDITIONS



VVI • 30 HIGH-YIELD TOPICS

MBBS | BHMS | B.Sc. NURSING | BDS | BPT

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VVI EXAM REVISION • NEURO PHYSIOTHERAPY

TRAUMATIC BRAIN INJURY
ASSESSMENT & REHABILITATION

Early Recognition Better Outcomes ♡

Topic - 5

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1. Introduction

- TBI is an injury to the brain due to an external mechanical force.
- Can be mild, moderate or severe.
- Leading cause of disability & death (especially in young adults).
- Common causes: road traffic accidents, falls, assaults, sports, violence.

2. Types of TBI

A. Closed Head Injury
- No skull fracture.
- Brain injury due to acceleration/ deceleration.

B. Open Head Injury (Penetrating)
- Skull fracture with brain involvement.
- High risk of infection.

C. By Severity (GCS)
- Mild : 13-15
- Moderate : 9-12
- Severe : 3-8

3. Clinical Features

A. Initial (Acute)
- Loss of consciousness
- Confusion / disorientation
- Headache, vomiting
- Seizures (early)
- Focal neurological deficits
- Skull fracture signs (raccoon eyes, battle sign, CSF leak)

B. Late (Post-Acute / Chronic)
- Cognitive deficits (memory, attention)
- Behavioural changes
- Motor & sensory deficits
- Speech and language problems
- Emotional and psychological issues

4. Assessment

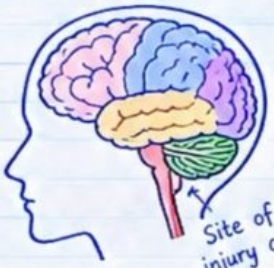
A. History
- Mechanism of injury
- Loss of consciousness (duration)
- Amnesia (retrograde / anterograde)
- Symptoms (headache, vomiting, seizure)
- Past medical history & comorbidities

B. Physical Examination
- General examination (vitals, trauma signs)
- Neurological examination (GCS, pupil, motor, sensory, reflexes)
- Cranial nerve assessment
- Signs of raised ICP (Cushing's triad)

C. Investigations
- CT scan (first line – detects bleed, fracture, mass effect)
- MRI (better for diffuse axonal injury)
- EEG (seizures)
- Neuropsychological tests (cognition, memory)
- X-ray (skull / cervical spine if needed)

GCS Components
- Eye opening (1-4)
- Verbal response (1-5)
- Motor response (1-6)

Total = 3 - 15



5. Management (Acute Phase)

A. Medical
- Airway, breathing, circulation (ABC)
- Control of raised ICP (head elevation, mannitol, hypertonic saline)
- Seizure prophylaxis (e.g., levetiracetam)
- Analgesics, antiemetics
- Treat associated injuries

B. Surgical
- Evacuation of hematoma
- Decompressive craniectomy (if needed)
- Repair of skull fracture / dural tear

⚠ Red Flags (Need urgent action)
- GCS < 8
- ✓ Pupillary changes
- ✓ Rapid deterioration
- ✓ Expending hematoma
- ✓ Signs of brain herniation

6. Rehabilitation

A. Early Phase (ICU / Hospital)
- Prevent complications (bed sores, DVT, contractures)
- Positioning & early mobilization
- Respiratory care
- Nutrition & hydration
- Family education

B. Later Phase
- Physical therapy (mobility, balance, strength)
- Occupational therapy (ADL training)
- Speech & language therapy (communication)
- Cognitive rehabilitation (memory, attention)

C. Goals
- ✓ Maximize functional independence
- ✓ Improve quality of life
- ✓ Reintegration into daily, academic & social life

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Small steps lead to big progress ♡

THANK YOU FOR YOUR PURCHASE!

We wish you the very best in your medical and physiotherapy examinations.

For any questions, supplementary questions, or study support, reach out to us:

SUPPORT & COMMUNITY:

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